Diagonalization

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Theorem 23.1: Let $\mathbf{A}$ be a triangular matrix (either upper or lower). Then the eigenvalues of $\mathbf{A}$ are its diagonal entries.

Definition. An $n \times n$ matrix A is **diagonalizable** if it is similar to a diagonal matrix: that is, if there exists an invertible $n \times n$ matrix C and a diagonal matrix D such that

$$A = CDC^{-1}.$$

Example. Any diagonal matrix is D is diagonalizable because it is similar to itself. For instance,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = I_3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} I_3^{-1}.$$

Example. Most of the examples in Section 5.3 involve diagonalizable matrices:

$$\begin{pmatrix} -12 & 15 \\ -10 & 13 \end{pmatrix} \text{ is diagonalizable because it equals } \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$\begin{pmatrix} 1/2 & 3/2 \\ 3/2 & 1/2 \end{pmatrix} \text{ is diagonalizable because it equals } \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$\frac{1}{5}\begin{pmatrix} -8 & -9 \\ 6 & 13 \end{pmatrix} \text{ is diagonalizable because it equals } \frac{1}{2}\begin{pmatrix} -1 & -3 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}\left(\frac{1}{2}\begin{pmatrix} -1 & -3 \\ 2 & 1 \end{pmatrix}\right)^{-1}$$

$$\begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix} \text{ is diagonalizable because it equals } \begin{pmatrix} -1 & 1 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} -1 & 1 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}^{-1}$$

Powers of diagonalizable matrices Multiplying diagonal matrices together just multiplies their diagonal entries:

$$\begin{pmatrix} x_1 & 0 & 0 \\ 0 & x_2 & 0 \\ 0 & 0 & x_3 \end{pmatrix} \begin{pmatrix} y_1 & 0 & 0 \\ 0 & y_2 & 0 \\ 0 & 0 & y_3 \end{pmatrix} = \begin{pmatrix} x_1 y_1 & 0 & 0 \\ 0 & x_2 y_2 & 0 \\ 0 & 0 & x_3 y_3 \end{pmatrix}.$$

Therefore, it is easy to take powers of a diagonal matrix:

$$\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}^n = \begin{pmatrix} x^n & 0 & 0 \\ 0 & y^n & 0 \\ 0 & 0 & z^n \end{pmatrix}.$$

By this fact in Section 5.3, if $A = CDC^{-1}$ then $A^n = CD^nC^{-1}$, so it is also easy to take powers of *diagonalizable* matrices. This will be very important in applications to difference equations in Section 5.6.

Recipe: Compute powers of a diagonalizable matrix. If $A = CDC^{-1}$, where D is a diagonal matrix, then $A^n = CD^nC^{-1}$:

$$A = C \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix} C^{-1} \quad \Longrightarrow \quad A^n = C \begin{pmatrix} x^n & 0 & 0 \\ 0 & y^n & 0 \\ 0 & 0 & z^n \end{pmatrix} C^{-1}.$$

Example. Let

$$A = \begin{pmatrix} 1/2 & 3/2 \\ 3/2 & 1/2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}.$$

Find a formula for A^n in which the entries are functions of n, where n is any positive whole number.

Solution. We have

$$\begin{aligned} A^n &= \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}^n \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-1)^n \end{pmatrix} \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2^n & (-1)^n \\ 2^n & (-1)^{n+1} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 2^n + (-1)^n & 2^n + (-1)^{n+1} \\ 2^n + (-1)^{n+1} & 2^n + (-1)^n \end{pmatrix}, \end{aligned}$$

Conditions for Diagonalization

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Easy Example. Apply the diagonalization theorem to the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 3 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$$

Example 23.8. Determine if $\mathbf{A}$ is diagonalizable. If yes, find a matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution. The characteristic polynomial of $\mathbf{A}$ is

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 2)(\lambda - 3)(\lambda + 1)$$

and therefore $\lambda_1 = 2$, $\lambda_2 = 3$, and $\lambda_3 = -1$ are the eigenvalues of $\mathbf{A}$. Since $\mathbf{A}$ has $n = 3$ distinct eigenvalues, then by Theorem 23.6 $\mathbf{A}$ is diagonalizable. Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ corresponding to $\lambda_1, \lambda_2, \lambda_3$ are found to be

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Therefore, a matrix that diagonalizes $\mathbf{A}$ is

$$\mathbf{P} = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

You can verify that

$$\mathbf{P} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathbf{P}^{-1} = \mathbf{A}$$

□

Example 23.9. Determine if $\mathbf{A}$ is diagonalizable. If yes, find a matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

Solution. The characteristic polynomial of $\mathbf{A}$ is

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 1)(\lambda - 2)^2$$

and therefore $\lambda_1 = 1$, $\lambda_2 = 2$. An eigenvector corresponding to $\lambda_1 = 1$ is

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

One finds that $g_2 = \dim(\mathrm{Null}(\mathbf{A} - \lambda_2\mathbf{I})) = 2$, and two linearly independent eigenvectors for λ_2 are

$$\{\mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Therefore, $\mathbf{A}$ is diagonalizable, and a matrix that diagonalizes $\mathbf{A}$ is

$$\mathbf{P} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -2 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

You can verify that

$$\mathbf{P} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathbf{P}^{-1} = \mathbf{A}$$

□

Example 23.10. Determine if $\mathbf{A}$ is diagonalizable. If yes, find a matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

Solution. The characteristic polynomial of $\mathbf{A}$ is

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$$

and therefore the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 1$ and $\lambda_2 = -2$. For $\lambda_2 = -2$ one computes

$$\mathbf{A} - \lambda_2\mathbf{I} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

We see that the dimension of the eigenspace of $\lambda_2 = -2$ is $g_2 = 1$, which is less than the algebraic multiplicity $k_2 = 2$. Therefore, from Theorem 23.7 we can conclude that it is not possible to construct a basis of eigenvectors of $\mathbf{A}$, and therefore $\mathbf{A}$ is not diagonalizable. □